



Danish Energy Agency

# Presentation

23/10/2025





Danish Energy Agency

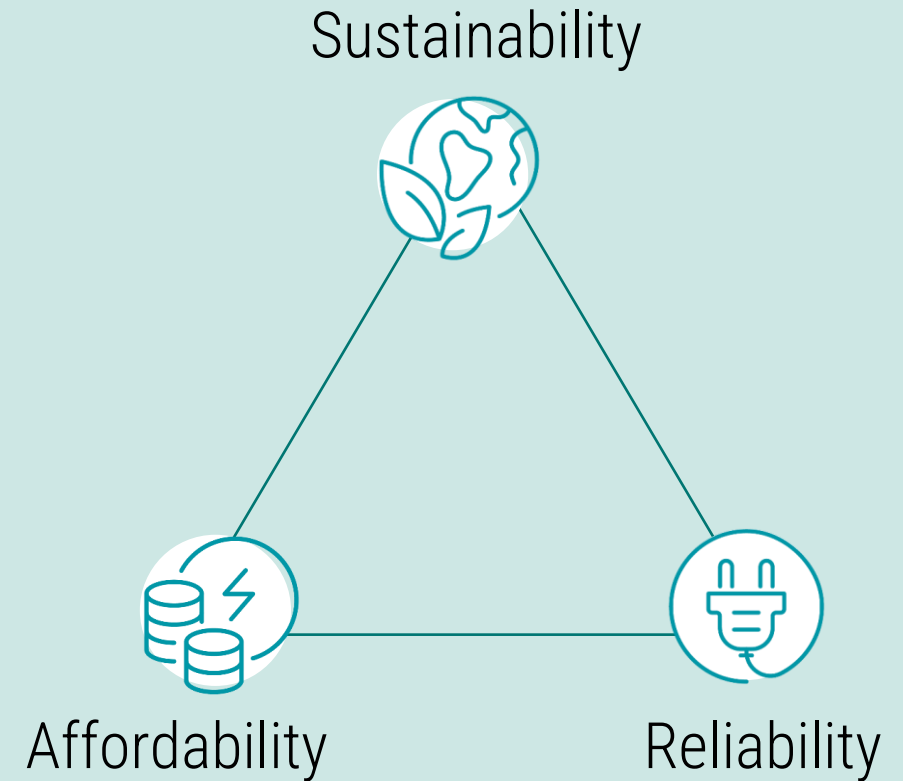
# Agenda

1. **Welcome to DEA**
2. **Data and Statistics (15 min.)**
3. **NECP (15 min.)**
4. **Spotlight on Denmark's Work with Bioenergy (15 min.)**
5. **Questions and Closing**

23/10/2025

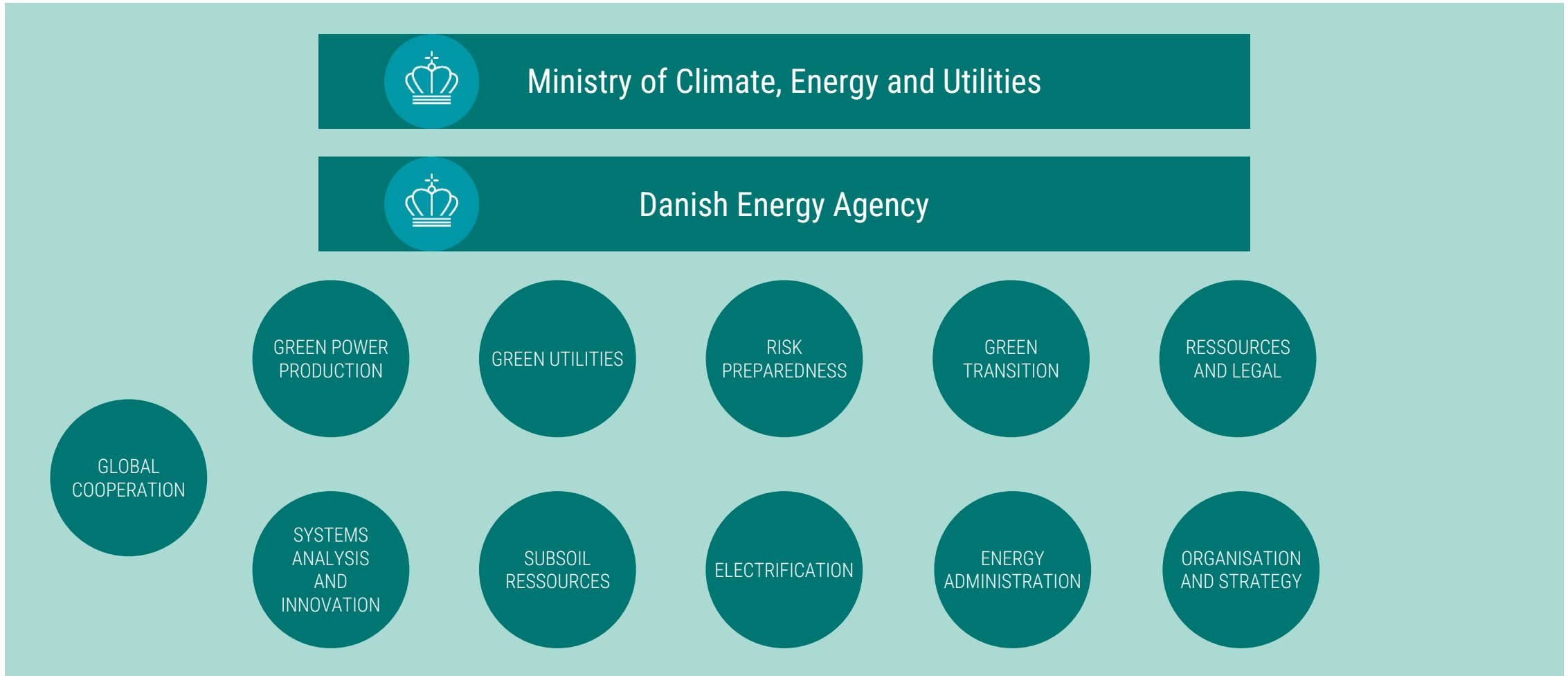


DEA work to overcome the energy trilemma - energy that is sustainable/green, secure and affordable.



# The Danish Energy Agency (ADE)

*Part of the Ministry of Climate, Energy and Utilities*



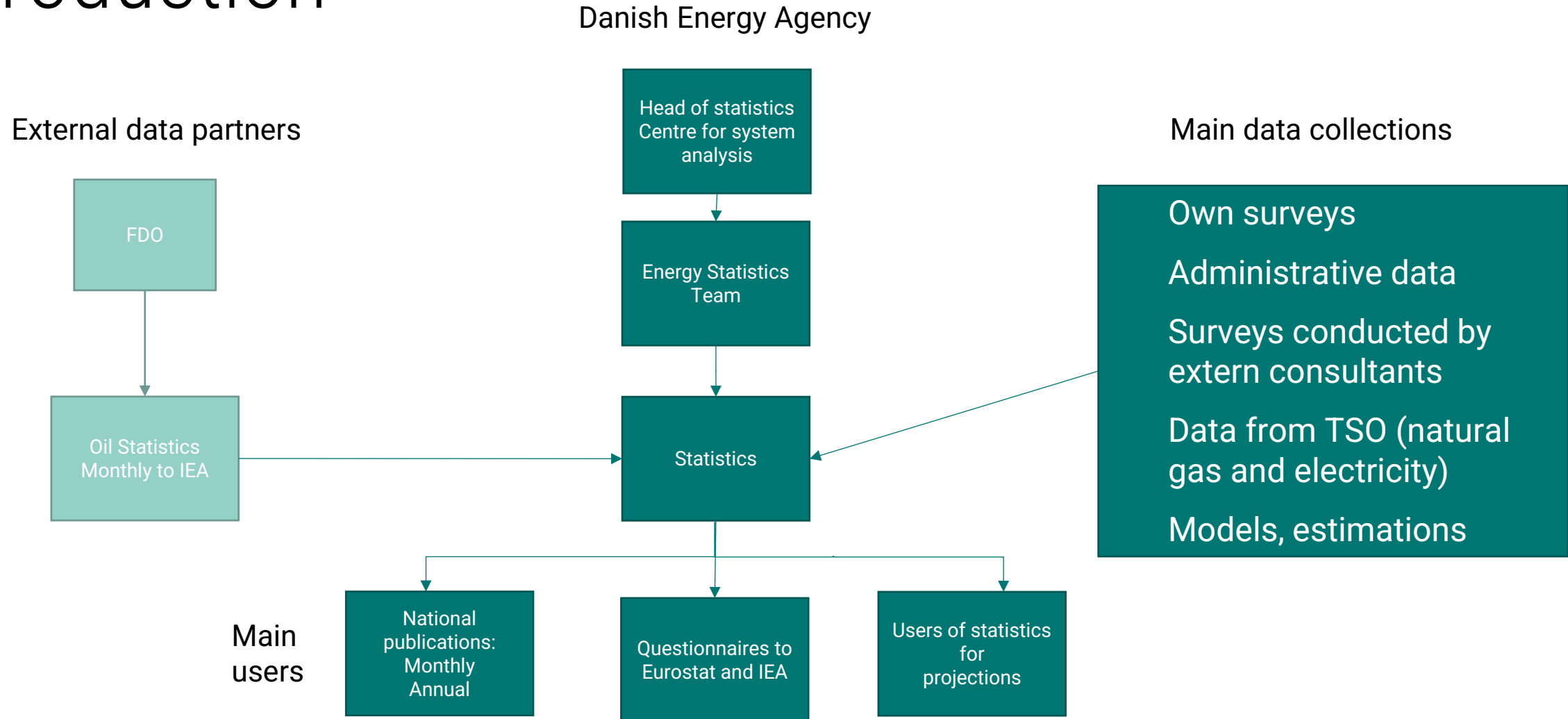
# The Danish Energy Statistics

- Institutional setup and legal framework
- Annual plan for Energy statistics compilation and main data sources
- Dissemination

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# Introduction

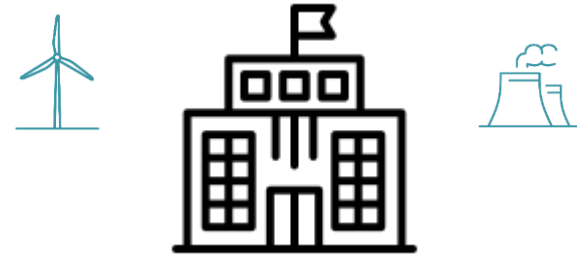


# Organization



Statistics Denmark - DST  
(National Statistical Institute)

➤ **statistical knowledge**



Danish Energy Agency - DEA  
(also a national authority)

- **close to the energy data**
- **understanding of the energy system:  
validation of data developments**

# When producing official statistics...

- Data collected for statistical purposes may only be used for statistics compilation
- Data is published in an aggregated form, so confidentiality is ensured (unless it is agreed otherwise with the data sources)
- Everybody get access to the data at the same time
- Publication dates are announced in advance
- Transparency about the data sources and the methodology





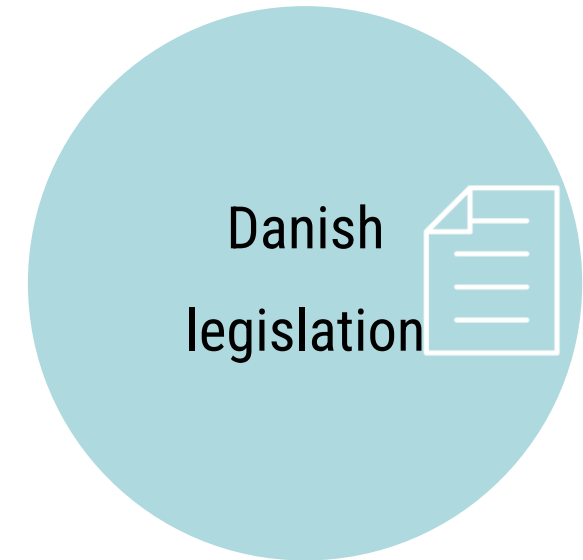
# Manuals and Legal framework



Energy Statistics Manual 2004

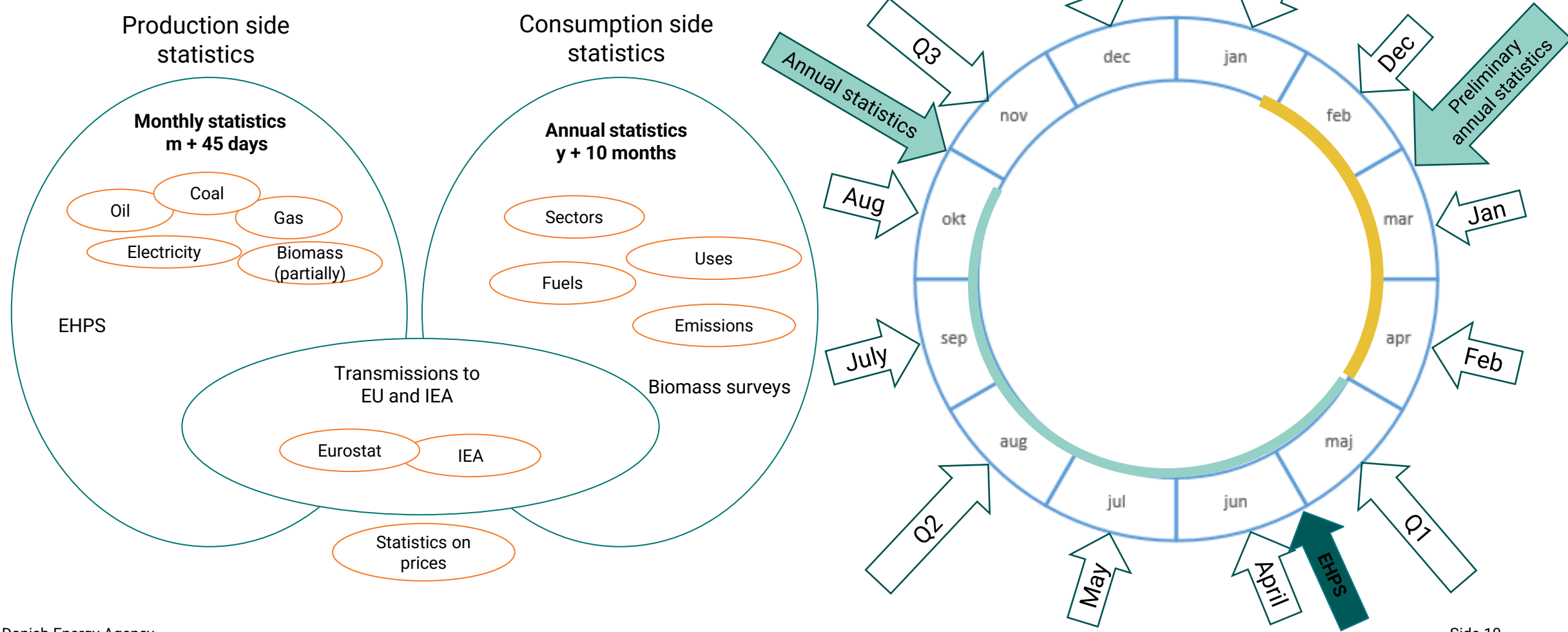


Statistics Regulation  
Code of Practice  
Energy Statistics Regulation  
Renewable energy directive

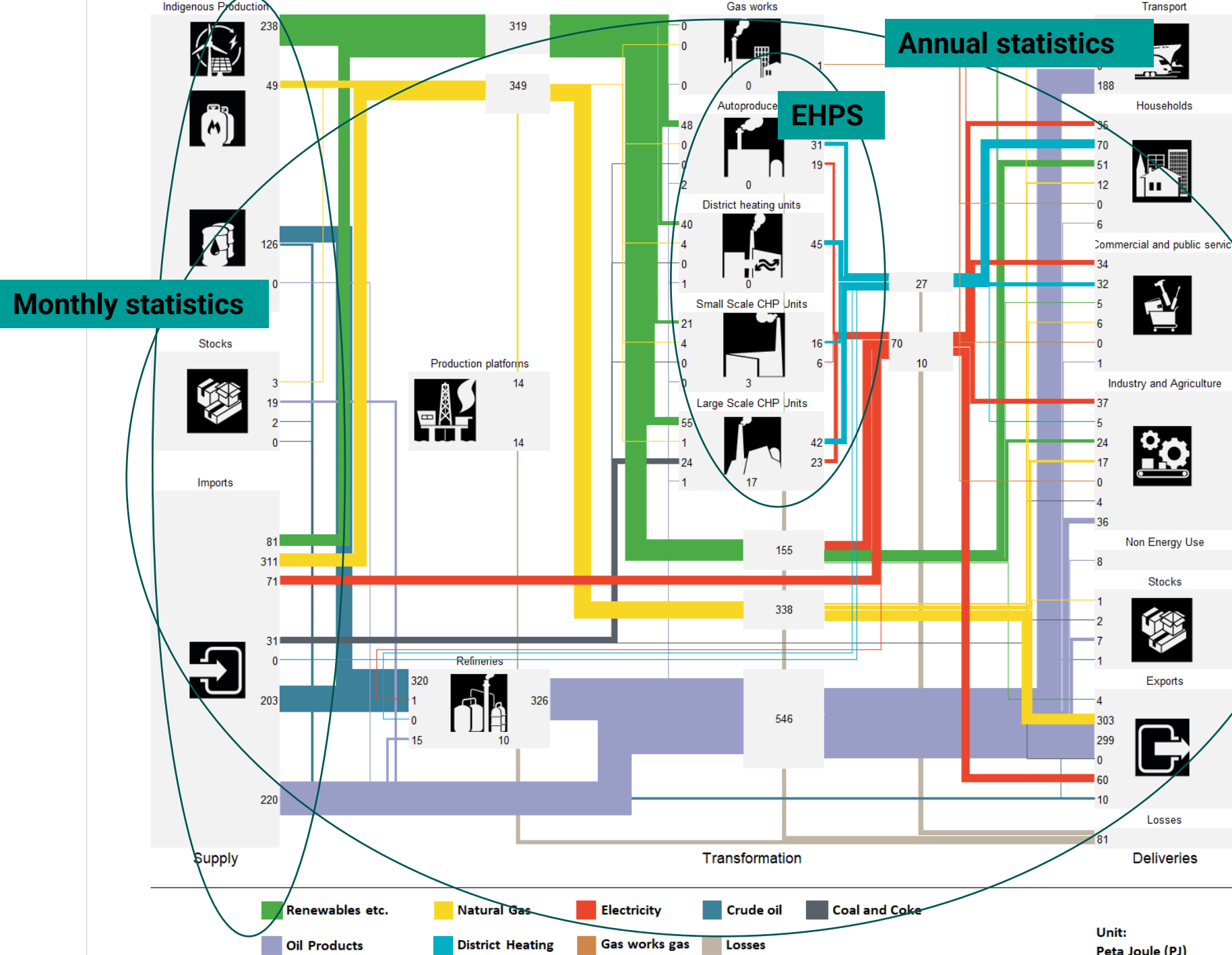


Act on oil stocks  
Act on natural gas supply  
Act on electricity supply  
Act on heat supply  
Act on hydrogen data

# Annual plan for energy statistics



# Danish Energy Flows 2023



Our website: <https://ens.dk/en/analyses-and-statistics/annual-and-monthly-statistics>

## Monthly Energy Statistics

Danish Energy Agency has published [monthly energy production and consumption statistics](#), which are available online in excel format. (Latest version: November 2024. Next version for December 2024 will be available March 14<sup>th</sup> 2025).

- **Oil Supply** 

Since January 2005, the Danish Energy Agency has published a monthly oil supply statistics. (Latest version: November 2024. Next version for December 2024 will be available February 13<sup>th</sup> 2025).

- **Electricity** 

Since January 2000, the Danish Energy Agency has published a monthly electricity supply statistics. (Latest version: November 2024. Next version for December 2024 will be available February 13<sup>th</sup> 2025).

- **Coal and Coke Supply** 

Since January 2001, the Danish Energy Agency has published a monthly coal and coke supply statistics. (Latest version: November 2024. Next version for December 2024 will be available February 13<sup>th</sup> 2025).

- **Natural Gas** 

The Danish Energy Agency prepares monthly natural gas supply statistics, which are available online in excel format. (Latest version: November 2024. Next version for December 2024 will be available February 13<sup>th</sup> 2025).

## Energy statistics 2023

### Energy Statistics 2023

[Download the Diagrams from the publication](#) 

Figures, tables and time series from the publication are available as MS Excel files below:

[Figures 2023](#) 

[Tables 2023](#) 

[Basic Data 2023](#) 

# Our main publication



Energistytelsen

# 2023

Data, tables, statistics and maps  
Energy Statistics 2023

## ENERGY BALANCE 2023

### Energy balance 2023

|                                              | Total           | Crude oil and refinery feedstocks | Oil products    | Natural gas    | Coal and coke  | Waste, non-renewable | Renewable energy | Electricity     | District heating |
|----------------------------------------------|-----------------|-----------------------------------|-----------------|----------------|----------------|----------------------|------------------|-----------------|------------------|
| <b>Direct energy content [TJ]</b>            |                 |                                   |                 |                |                |                      |                  |                 |                  |
| Total energy consumption                     | 566 926         | 317 183                           | -63 901         | 58 363         | 27 930         | 17 349               | 298 608          | 11 280          | 1                |
| Primary energy production                    | 412 669         | 125 647                           | -               | 48 642         | -              | 15 154               | 223 226          | -               | -                |
| Recycling                                    | 1               | -                                 | 1               | -              | -              | -                    | -                | -               | -                |
| Imports                                      | 915 875         | 205 251                           | 216 986         | 310 513        | 30 555         | 2 195                | 78 871           | 71 391          | 1                |
| Exports                                      | -643 253        | -12 893                           | -263 014        | -303 054       | -231           | -                    | -3 950           | -60 111         | -                |
| Border trade                                 | -8 045          | -                                 | -8 045          | -              | -              | -                    | -                | -               | -                |
| International marine bunkers                 | -22 910         | -                                 | -22 910         | -              | -              | -                    | -                | -               | -                |
| Stock changes                                | 11 251          | 2 126                             | 11 790          | -833           | -2 426         | -                    | 594              | -               | -                |
| Statistical differences, input from blending | 1 340           | -2 948                            | 1 292           | 3 097          | 32             | -                    | -132             | -               | -                |
| <b>Energy sector</b>                         | <b>-27 498</b>  | <b>-317 183</b>                   | <b>307 737</b>  | <b>-13 505</b> | <b>-</b>       | <b>-</b>             | <b>-</b>         | <b>-4 540</b>   | <b>-</b>         |
| Extraction and gasification                  | -13 505         | -                                 | -               | -13 505        | -              | -                    | -                | -               | -                |
| Petroleum products                           | 322 606         | -                                 | 322 606         | -              | -              | -                    | -                | -               | -                |
| Used in refineries                           | -333 103        | -317 183                          | -14 870         | -              | -              | -                    | -                | -1 043          | -                |
| Used in distribution                         | -3 497          | -                                 | -               | -              | -              | -                    | -                | -3 497          | -                |
| <b>Transformation</b>                        | <b>-20 089</b>  | <b>-</b>                          | <b>-3 621</b>   | <b>-10 393</b> | <b>-24 355</b> | <b>-15 982</b>       | <b>-209 829</b>  | <b>109 690</b>  | <b>133 8</b>     |
| Large-scale units                            | -14 614         | -                                 | -696            | -1 285         | -24 355        | -                    | -55 441          | 25 513          | 41 6             |
| Wind turbines and hydropower plants          | -               | -                                 | -               | -              | -              | -                    | -69 886          | 69 886          | -                |
| Small-scale units                            | -1 693          | -                                 | -50             | -4 016         | -              | -2 288               | -18 504          | 6 506           | 16 6             |
| District heating units                       | -160            | -                                 | -746            | -4 419         | -              | -230                 | -31 258          | -8 729          | 45 2             |
| Autoproducers                                | 163             | -                                 | -2 129          | -379           | -              | -13 464              | -34 439          | 19 534          | 31 6             |
| Gas works                                    | -87             | -                                 | -0              | -295           | -              | -                    | -300             | -               | -                |
| Own use                                      | -3 698          | -                                 | -               | -              | -              | -                    | -                | -3 020          | -6               |
| <b>Distribution losses etc.</b>              | <b>-33 880</b>  | <b>-</b>                          | <b>-</b>        | <b>-72</b>     | <b>-</b>       | <b>-</b>             | <b>-42</b>       | <b>-6 944</b>   | <b>-26 8</b>     |
| <b>Final energy consumption</b>              | <b>-585 458</b> | <b>-</b>                          | <b>-240 215</b> | <b>-34 393</b> | <b>-3 575</b>  | <b>-1 368</b>        | <b>-88 737</b>   | <b>-109 486</b> | <b>-107 1</b>    |
| Non-energy use                               | -7 947          | -                                 | -7 947          | -              | -              | -                    | -                | -               | -                |
| Transport                                    | -201 296        | -                                 | -188 226        | -304           | -              | -                    | -9 238           | -3 528          | -                |
| Agriculture and industry                     | -123 554        | -                                 | -36 417         | -16 729        | -3 575         | -1 154               | -23 154          | -36 872         | -5 467           |
| Commercial and public services               | -78 760         | -                                 | -1 379          | -5 792         | -              | -214                 | -5 212           | -33 979         | -32 161          |
| Households                                   | -173 901        | -                                 | -6 245          | -11 568        | -              | -                    | -51 133          | -35 108         | -69 570          |

Note: The energy balance provides an overview of supply (production, imports and exports), transformation and consumption of energy. A more detailed statement of input (black figures) and output (red figures) of energy products is listed in the table "Energy supply and consumption" table on pages 18-19.  
\*) LNG has been included with natural gas.

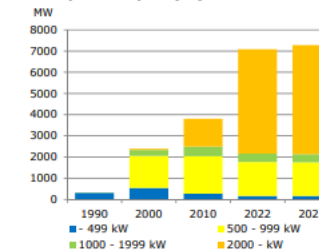
## RENEWABLE ENERGY

### Wind power - number of turbines and capacity by size

|                                | 1990    | 2000    |          |       | 2022    |          |       | 2023    |          |       |
|--------------------------------|---------|---------|----------|-------|---------|----------|-------|---------|----------|-------|
|                                | Onshore | Onshore | Offshore | Total | Onshore | Offshore | Total | Onshore | Offshore | Total |
| Total no. of turbines          | 2 666   | 6 194   | 41       | 6 235 | 5641    | 630      | 6271  | 5631    | 648      | 6279  |
| - 499 kW                       | 2 656   | 3 652   | 11       | 3 663 | 2152    | -        | 2152  | 2160    | -        | 2160  |
| 500 - 999 kW                   | 8       | 2 283   | 10       | 2 293 | 2346    | 10       | 2356  | 2318    | 10       | 2328  |
| 1 000 - 1 999 kW               | 2       | 251     | -        | 251   | 316     | -        | 316   | 316     | -        | 316   |
| 2 000 - kW                     | -       | 8       | 20       | 28    | 827     | 620      | 1447  | 837     | 628      | 1465  |
| Total wind power capacity [MW] | 326     | 2 340   | 50       | 2 390 | 4778    | 2306     | 7084  | 4808    | 2469     | 7277  |
| - 499 kW                       | 317     | 533     | 5        | 538   | 164     | -        | 164   | 162     | -        | 162   |
| 500 - 999 kW                   | 6       | 1 512   | 5        | 1 517 | 1600    | 5        | 1605  | 1580    | 5        | 1585  |
| 1 000 - 1 999 kW               | 3       | 279     | -        | 279   | 391     | -        | 391   | 391     | -        | 391   |
| 2 000 - kW                     | -       | 16      | 40       | 56    | 2623    | 2301     | 4924  | 2675    | 2464     | 5139  |

Note: The first offshore wind turbines were established in 1991.

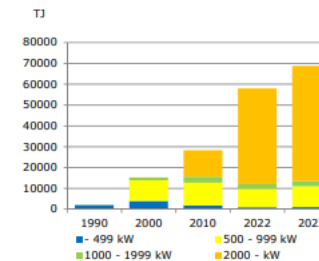
### Wind power capacity by size of turbine



The total number of wind turbines increased by 8 turbines from 2022 to 2023, and the total wind power capacity increased by 193 MW.

For years, the trend has been toward fewer but larger turbines. Since 2000, the number of small wind turbines of up to 499 kW has been reduced by 1,503, and the number of large turbines of more than 2 MW has increased by 1,447.

### Wind power production by size of turbine



The development toward larger turbines is even more evident in terms of wind power production. From 2000 to 2023, production from turbines up to 499 kW decreased by 2,915 TJ, while production from turbines of more than 2 MW increased by 56,598 TJ. However, from 2022 to 2023, capacity increased by 2.7%, while electricity production increased by 1.8%.

Turbines above 2 MW accounted for 70.6% of installed wind power capacity and were responsible for 80.9% of total energy from wind turbines in 2023.

Similarly, in 2023 wind turbines with a capacity up to 499 MW accounted for only 1.4% of the total production. Wind turbines with a capacity up to 499 kW accounted for 22.5% in 2000.

# Development of the NECP

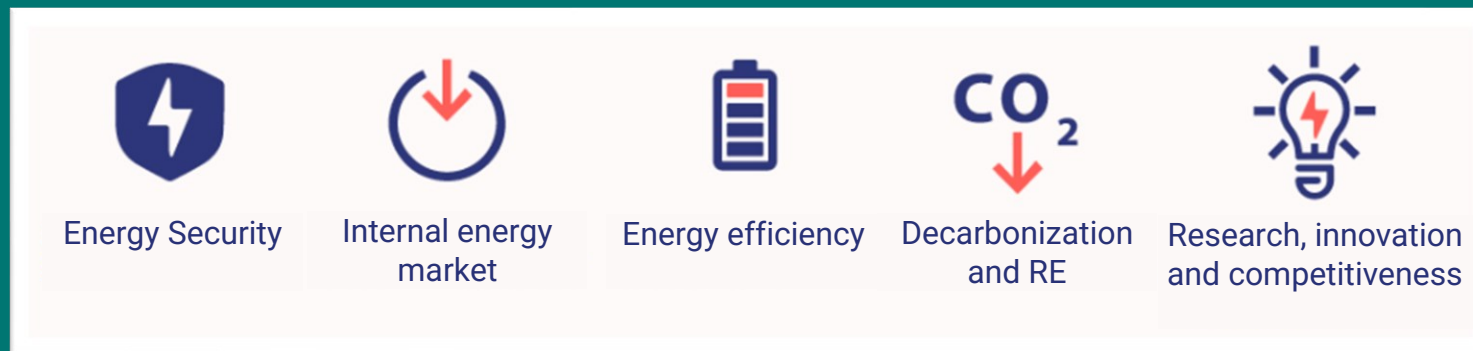
- *Description of the National Energy and Climate Plans (NECP)*
- *Framework for updating the NECP*
- *Practical work with NECP*
- *The interaction between the NECP and the EU's overall objectives*

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# Brief description of the NECP

- The National Energy and Climate Plans (NECPs) were introduced in 2016, as part of the EU Governance Regulation under *the Clean energy for all Europeans package*, which is part of the implementation of the Energy Union.
- The NECPs are the central reporting tool in the Governance Regulation and is intended to bring together most of the previous reporting obligations from, among others, the RE and EE directives and add new reporting in a comprehensive climate and energy policy plan.



# Purpose of the NECP

- NECPs from the Member States are used by the European Commission to assess whether the Member States are progressing on reaching their national targets and EU obligations, but more importantly, the Commission are using the NECPs to assess if EU as a whole are reaching the targets set in the Paris Agreement.
- The Commission makes recommendations for improvements and, based on the NECPs received, may also propose new EU legislation if the Member States' overall contribution is not ambitious enough to achieve the EU's objectives.





# NECP in Denmark

- Denmark submitted its first NECP in 2019, which was updated with a draft to the Commission in 2023 and the final version was delivered on 30<sup>th</sup> of June 2024.
- The framework for the update of the NECP in Denmark stemmed from the Governance Regulation, a series of meetings between the Commission and the Member States in the two years prior to the deadline, and the feedback the Commission gave on the draft NECPs they received in 2023.



# NECP in Denmark

- Due to developments in the EU's energy and climate policy and objectives, as well as COVID and the geopolitical developments that have occurred since DK submitted the first NECP in 2019, the Commission wanted additional content in the update of the NECP.

New political conditions since the first NECP

- *European Union*
  - European Green Deal and Climate Law
  - Fit for 55
  - REPowerEU
  - Recovery and Resilience Facility
  - Revised directives (RED, EED, EPBD)
- *Geopolitical developments*
  - War in Ukraine
  - Rising energy prices

# NECP in Denmark

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## Disclaimer

*Please note that this is a non-exhaustive and unofficial technical working document established based on the Commission Notice on the Guidance to Member States for the update of 2021-2030 national energy and climate plans (2022/C 495/02) following the request by some Member States for further clarity on the guidance and the corresponding sections in Annex I to the Governance Regulation. This document does not necessarily reflect the official opinion position of the Commission or of the Directorates- General for Energy and for Climate Action.*

*Text from guidance and its table*

## ANNEX I

### GENERAL FRAMEWORK FOR INTEGRATED NATIONAL ENERGY AND CLIMATE PLANS

#### Part 1

#### General framework

#### SECTION A: NATIONAL PLAN

#### 1. OVERVIEW AND PROCESS FOR ESTABLISHING THE PLAN

##### 1.1. Executive summary

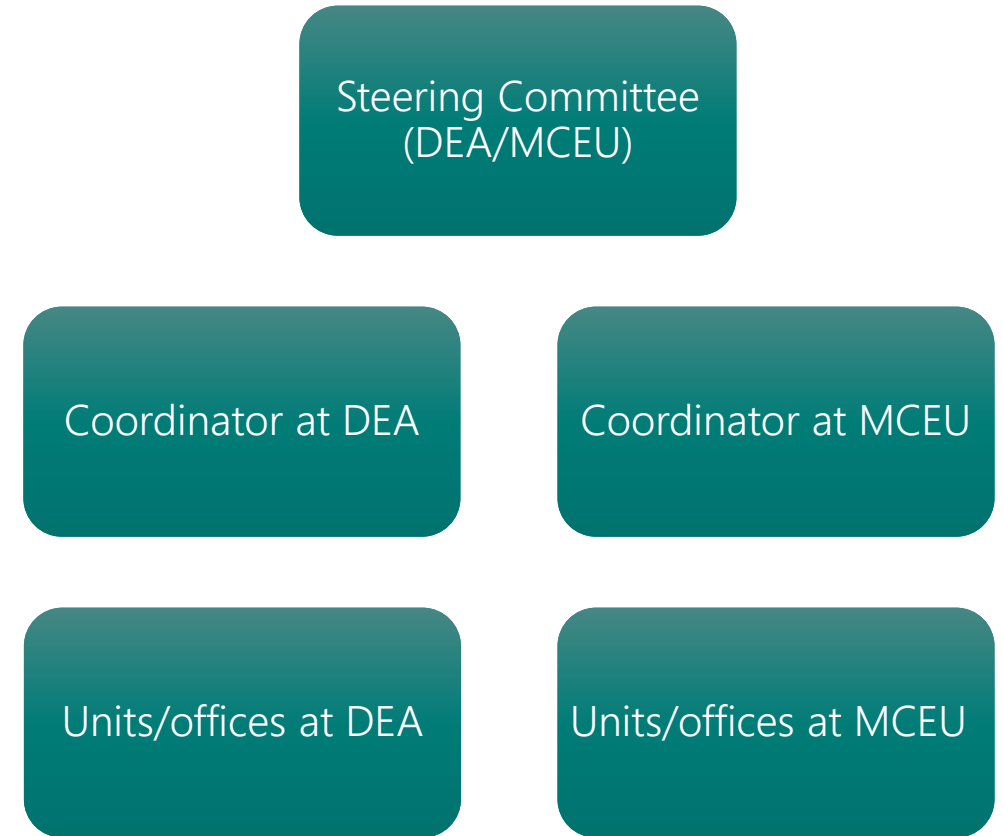
i. Political, economic, environmental, and social context of the plan.

*Member States to explain their national context vis a vis the below:*

- **Record high-energy prices** since the second half of 2021, exacerbated by Russia's unjustified and unlawful military aggression against Ukraine, give a strong impetus to accelerate the implementation of the European Green Deal and reinforce the resilience of the Energy Union by speeding up the clean energy transition and ending any dependence on Russian fossil fuels.
- At the same time, **recent extreme weather conditions** have put additional pressure on energy supplies, affected crop yields and inland navigation, and affected the wellbeing of citizens, giving a stark warning on the need to adapt to the impacts of climate change.
- Since the start of the energy price rise in mid-2021, the EU has adopted a comprehensive and tailored **emergency response**. This includes the need to diversify swiftly energy supply, to develop a consistent timetable for gas storage, to reflect the emergency demand reduction and emergency intervention to address high-energy prices and enhance solidarity.

# Project organization

- The work on the NECP is organized with a steering group including representatives from both our Ministry of Climate, Energy and Utilities (MCEU) and the Danish Energy Agency (DEA).
- The NECP includes a wide range of areas, so many of our colleagues from many different units/offices at both the DEA and MCEU, as well as other ministries and agencies, contribute with input and material to the plan. This could be both qualitative and quantitative material.
- In addition, meetings where also held with an external stakeholder group consisting of Danish actors, organizations, NGO's, etc. with an interest in the area.



# NECP in Denmark

- Due to the Danish Climate Law from 2019, we annually develop several reports, analyses, plans, etc. regarding climate and energy, that we use as the main inputs to our NECP.
- On top of this we also receive information and data from other ministries and agencies that we use as input to our NECP.

## Sources for the NECP include:

- Climate Status and Outlook
- Analysis Assumptions
- Climate Programme
- Energy statistics
- Material from other ministries
- Dialogue with relevant stakeholders in formal bodies and through consultations
- Etc.

# Scenario selection

- The NECP primarily uses data and projections from Climate Status and Outlook, which have been calculated on a frozen basis policy scenario, which is also the starting point of Article 3.2 (d) of the Governance Regulation.
- In the event that the Governance Regulation requested specific data sets and projections that were not included in our Climate Status and Outlook (CSO), those scenarios were modeled based on data from CSO, so that they specifically complied with the requirements of the Governance Regulation.
- Chapter 5 of the NECP provides an opportunity to account for the possible effects of “planned policies”, which are defined as policies and instruments that “...are under discussion and that have a realistic chance of being adopted and implemented after the date of submission.”. In the updated NECP, Denmark pointed to additional initiatives and made qualitative partial impact assessments, but no new scenarios were prepared. We did not include a quantitative scenario or assessment, because we did not have any data and we only want to include policies that are guaranteed to become a reality.

# Level of ambition in the Danish NECP

- The NECP itself does not set the level of ambition for energy and climate policy in Denmark.
- The NECP is used as a reporting tool in Denmark, where we report the Danish status to the Commission regarding how Denmark is reaching our EU obligations in the five dimensions that the NECP covers and therefore not a plan where we develop new policies and initiatives.
- In Denmark, the level of ambition is set by, among other things, the Danish Climate Law and EU directives, which are the primary governing tools for the development of energy and climate policy in Denmark.



Danish Energy Agency

# Denmark's Work with Bioenergy

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23/10/2025





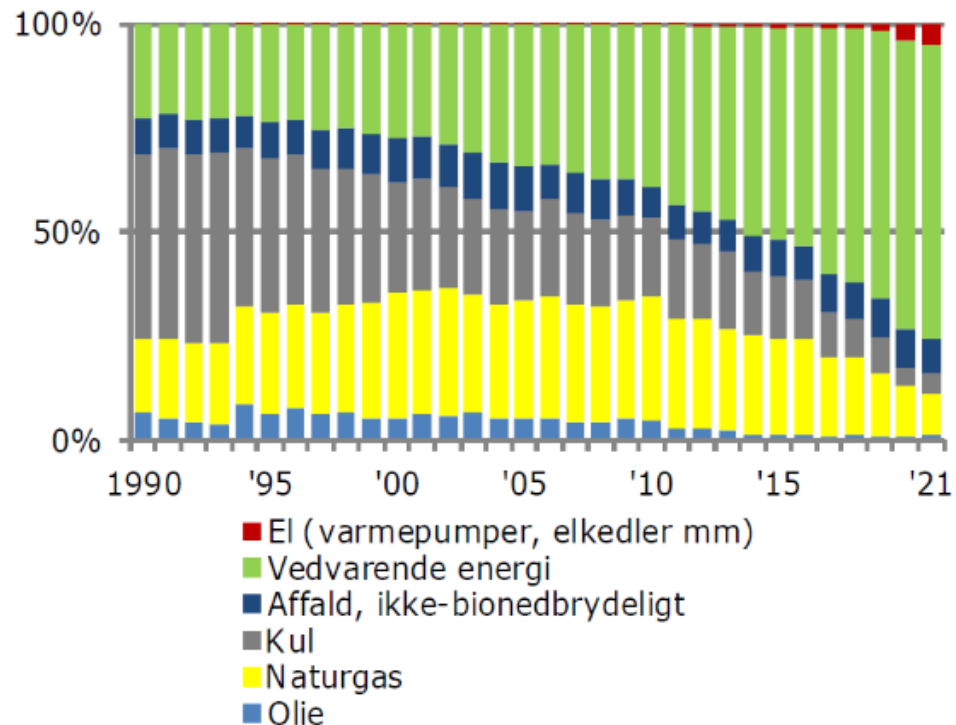
# The Danish case

- 5.9 mill. inhabitants
- 43 000 km<sub>2</sub>
- Highest elevation 171 m
- Forest 13 pct.
- Agricultural area 59 pct.
- Use of biomass fuels approx. 30 GJ / person

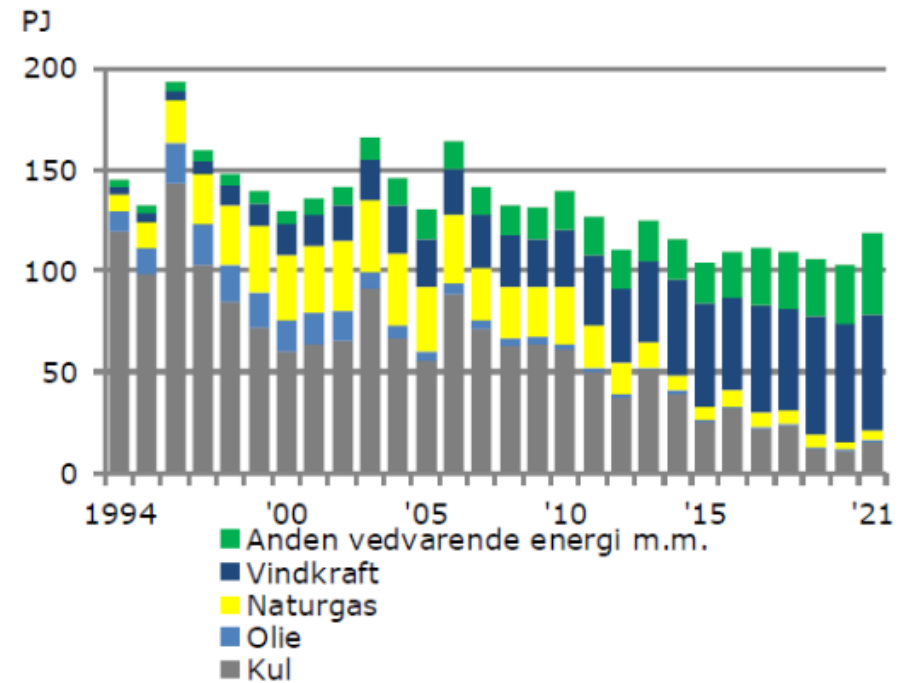


# A green transition of the energy system

Energy sources for district heating – biomass (green) has replaced coal (grey) and fossil gas (yellow)



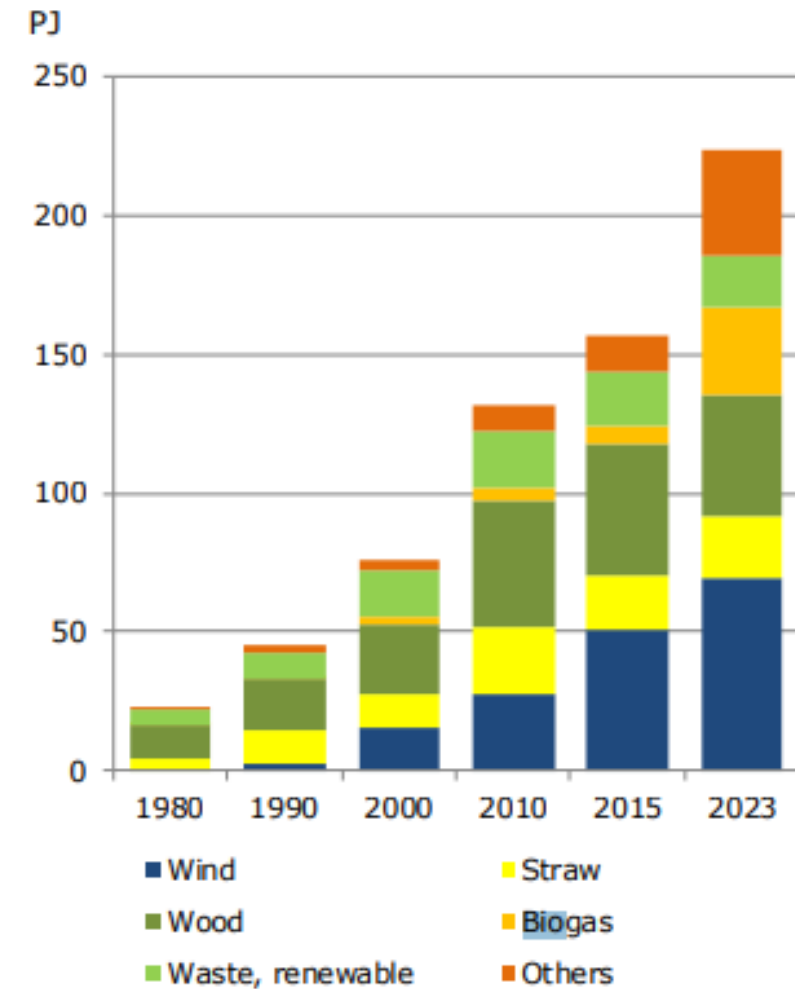
Energy sources for electricity production – windpower (dark blue) and biomass have replaced fossil gas and coal



# Key points on bioenergy in Denmark

- 2023: RE-production 223 PJ of a total of 413 PJ
- Biomass and RE-waste approx. 115 PJ
- Wood includes wood chips, firewood, wood pellets and wood waste

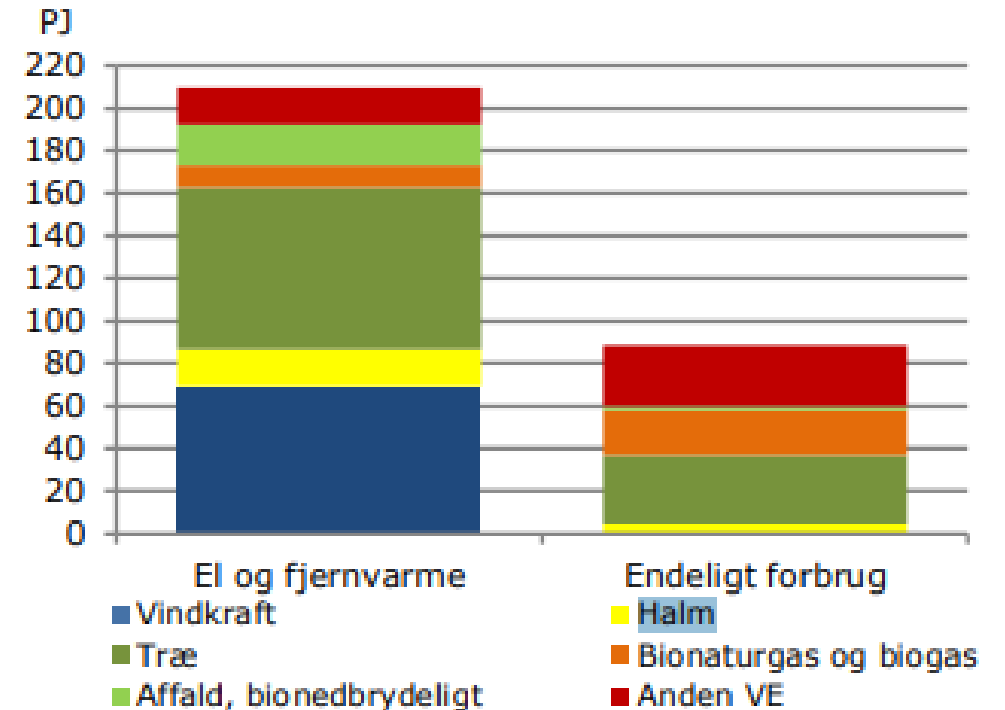
**Renewable energy production by type**



# Key points on bioenergy in Denmark

- Total consumption of RE just under 300 PJ in 2023
- Difference in production and consumption of woody biomass of approx. 60 PJ is due to import

Consumption of RE in 2023

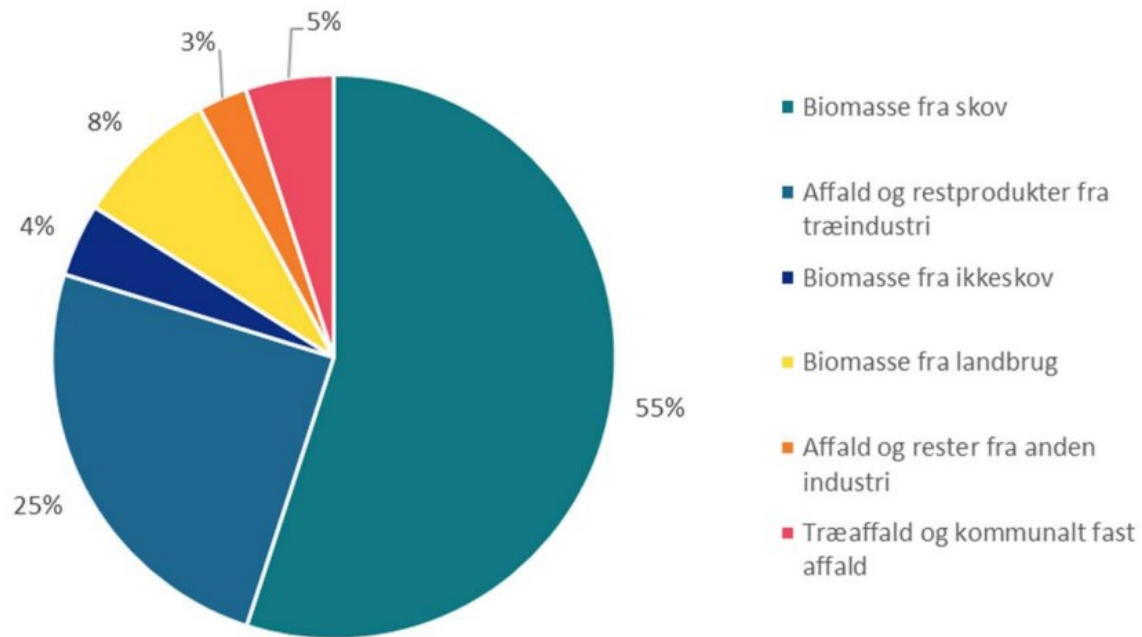


Biomass input

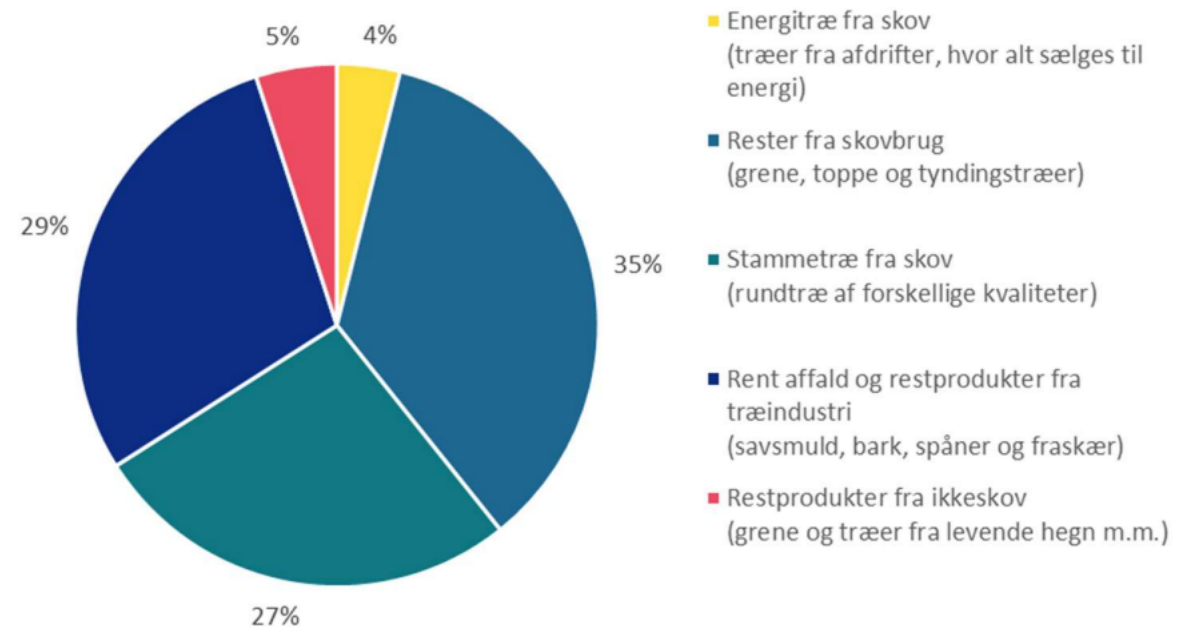


# Solid biomass input

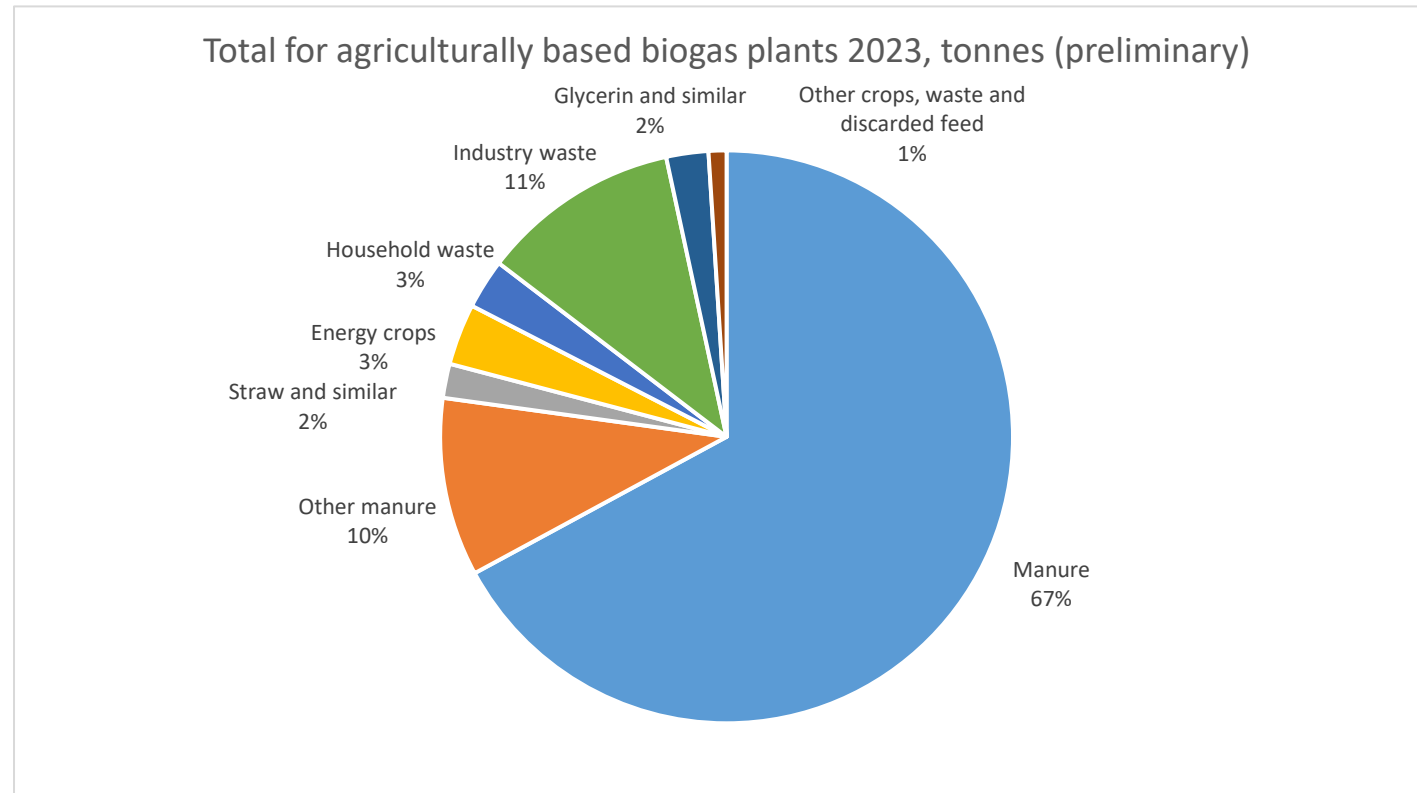
Figur 2: Biomassens sammensætning 2023 fordelt på biomassekategorier, totalt 7.119.300 ton.



Figur 3: Biomassetypers andel 2023 af den omfattede træbiomasse, totalt 5.973.700 ton.



# Input biomass for biogas



# Biomasses

| Biogas<br>2023                        | Tonnes     |       | Gas produced<br>mio. Nm3 |       |
|---------------------------------------|------------|-------|--------------------------|-------|
| Manure                                | 14.684.000 | 77,2% | 439                      | 35,2% |
| Energy Crops                          | 657.000    | 3,5%  | 101                      | 8,2%  |
| Crop residues                         | 232.000    | 1,2%  | 32                       | 2,6%  |
| Straw                                 | 260.000    | 1,4%  | 71                       | 5,8%  |
| Husk                                  | 108.000    | 0,6%  | 36                       | 3,0%  |
| Olive residues                        | 35.000     | 0,2%  | 10                       | 0,9%  |
| Potato/beet pulp                      | 453.000    | 2,4%  | 32                       | 2,6%  |
| Fisheries residues                    | 203.000    | 1,1%  | 29                       | 2,4%  |
| Slaugtherhouse waste                  | 434.000    | 2,3%  | 63                       | 5,1%  |
| Soapstock                             | 126.000    | 0,7%  | 48                       | 3,9%  |
| Melasse                               | 106.000    | 0,6%  | 32                       | 2,6%  |
| Glycerine                             | 452.000    | 2,4%  | 222                      | 17,8% |
| Industrial waste                      | 750.000    | 3,9%  | 63                       | 5,1%  |
| Household pulp                        | 530.000    | 2,8%  | 60                       | 4,9%  |
| Grand Total                           | 19.030.000 |       | 1.246                    |       |
| Energy content PJ                     |            |       | 29                       |       |
| Methane yield, Nm3 pr. tonnes biomass |            |       | 42,6                     |       |



# Challenges on input and sustainability

- Import of biomass
- Calculation of climate effect
- Energy crops
- Local opposition



# Bioenergy development and regulation



# Drivers for change

- No tax on biomass
- High energy tax on natural gas
- From coal to wood-fired powerplants
- Support schemes



# The Danish gas system

Increasing production of biogas and decreasing gas consumption

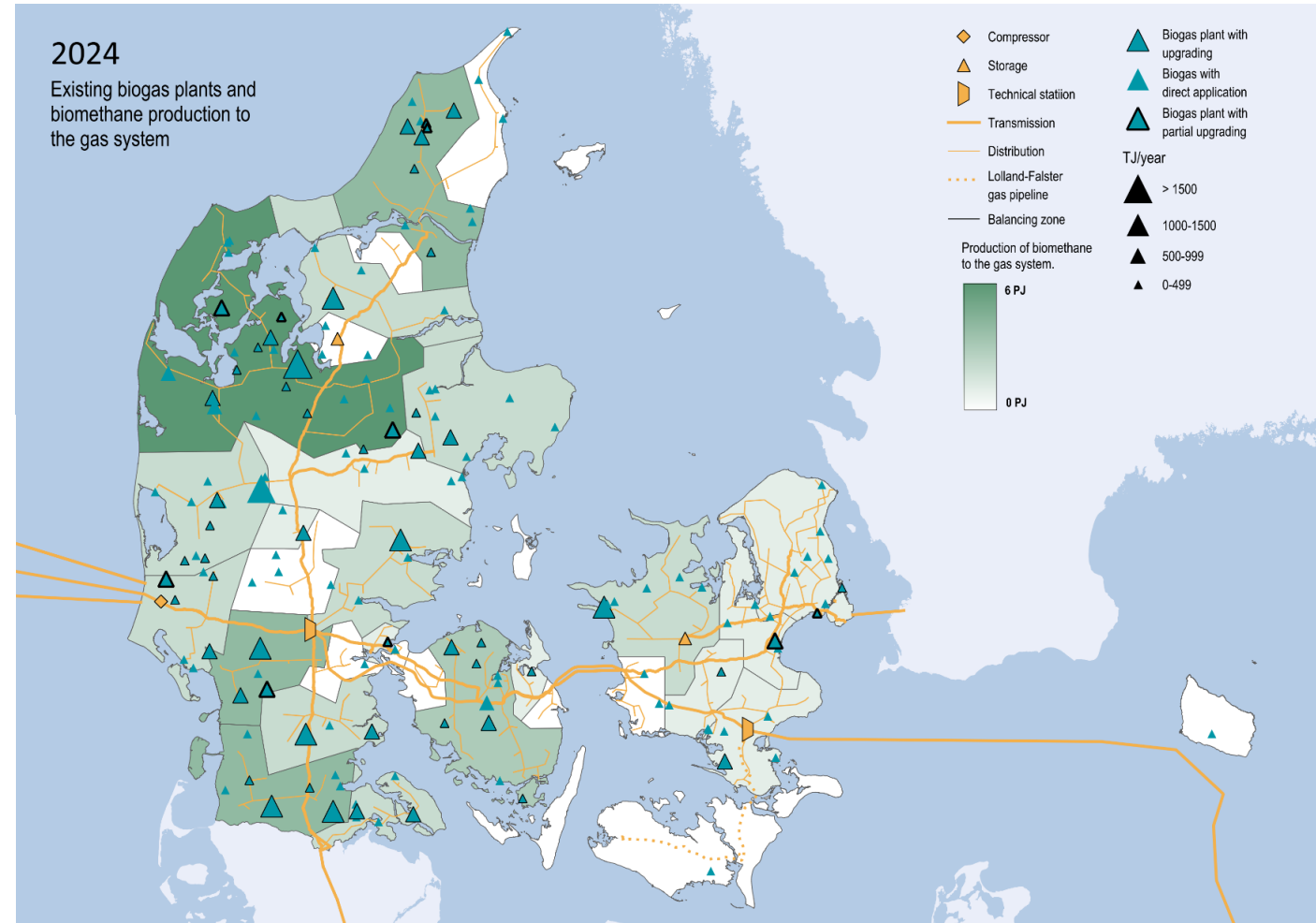
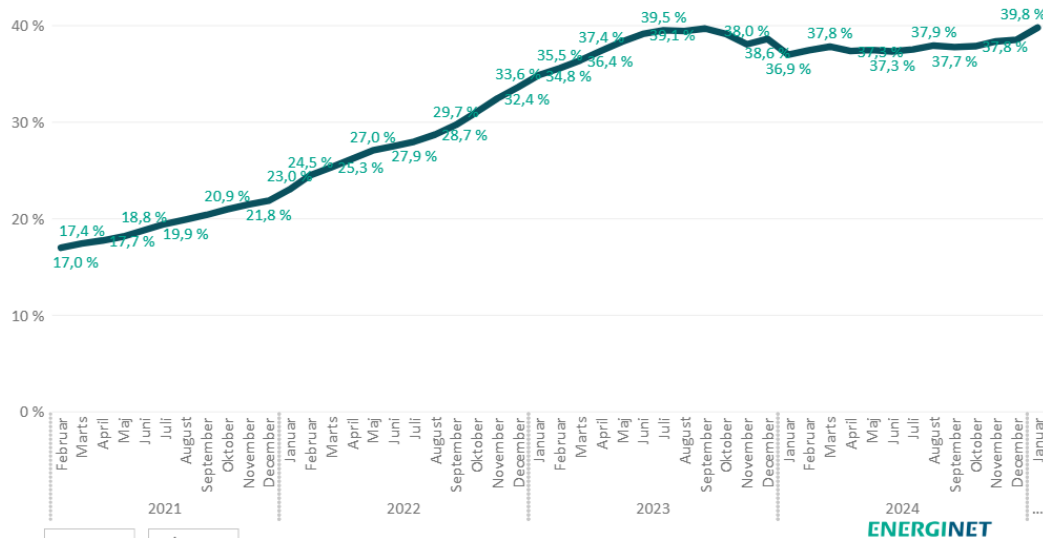
## Increasing production of biogas

Increase since 2014.

2024 ≈ 38 % of total consumption.

## Decreasing consumption

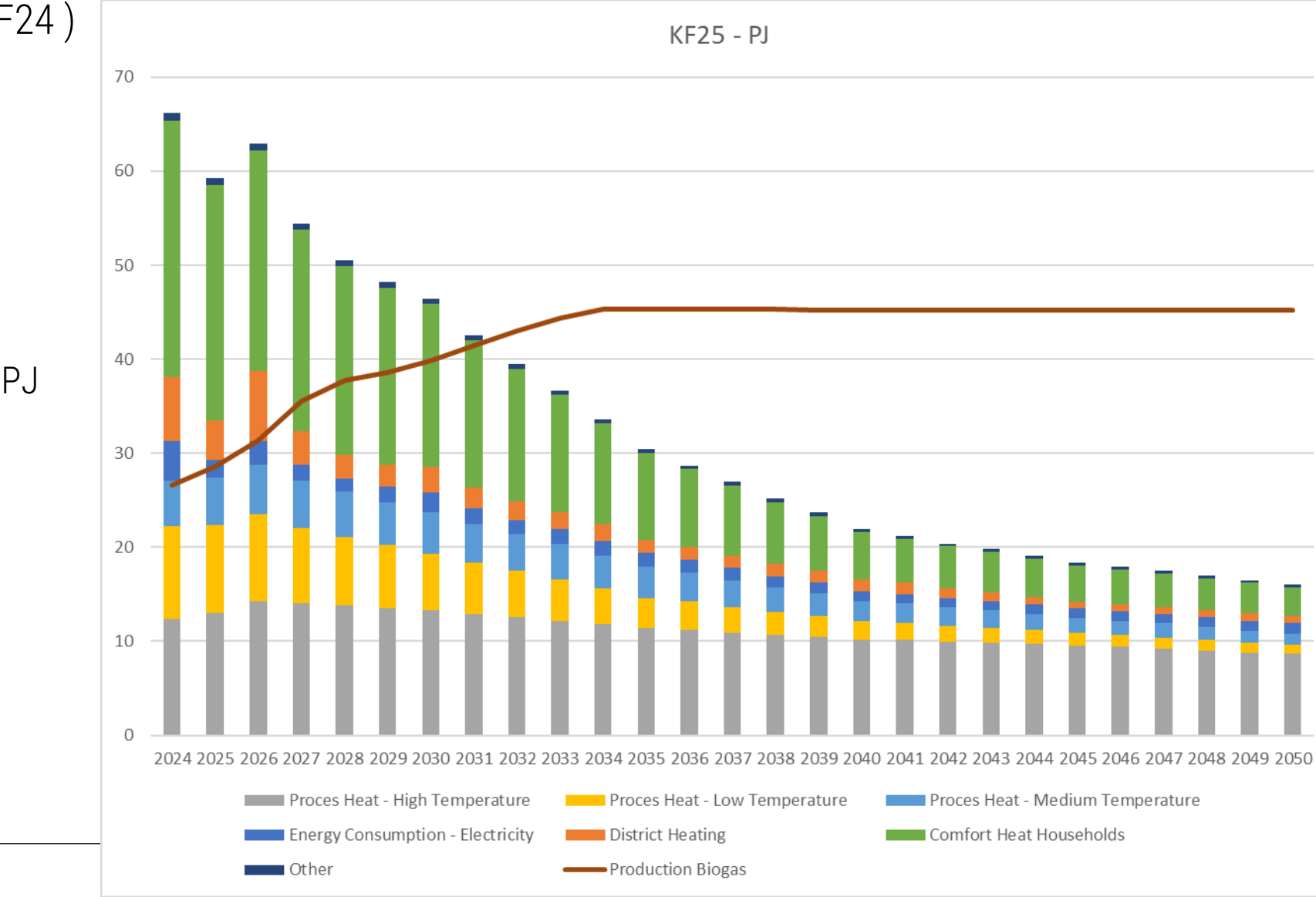
During the last 15 years gas consumption has decreased by 40% due to diminishing demand in gas for CHP and lately also households.





# Development in biogas production and consumption

Climate projection (KF24 )

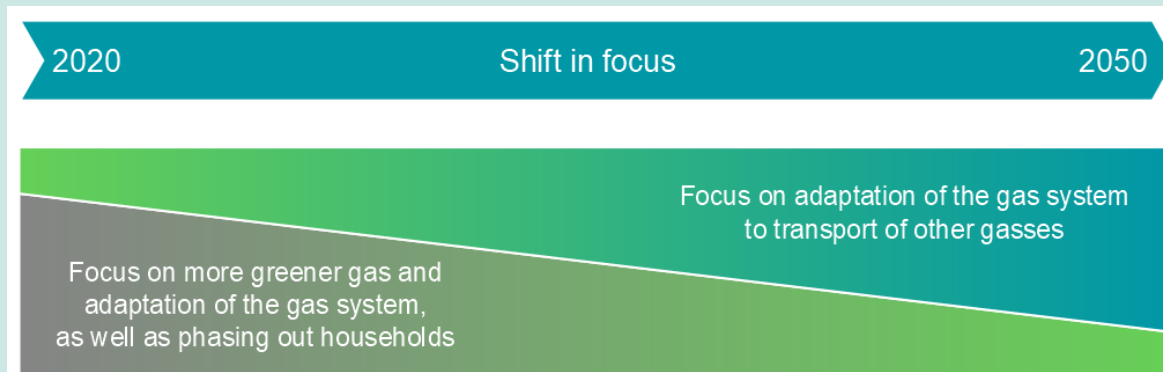


Challenges, politics  
and onwards

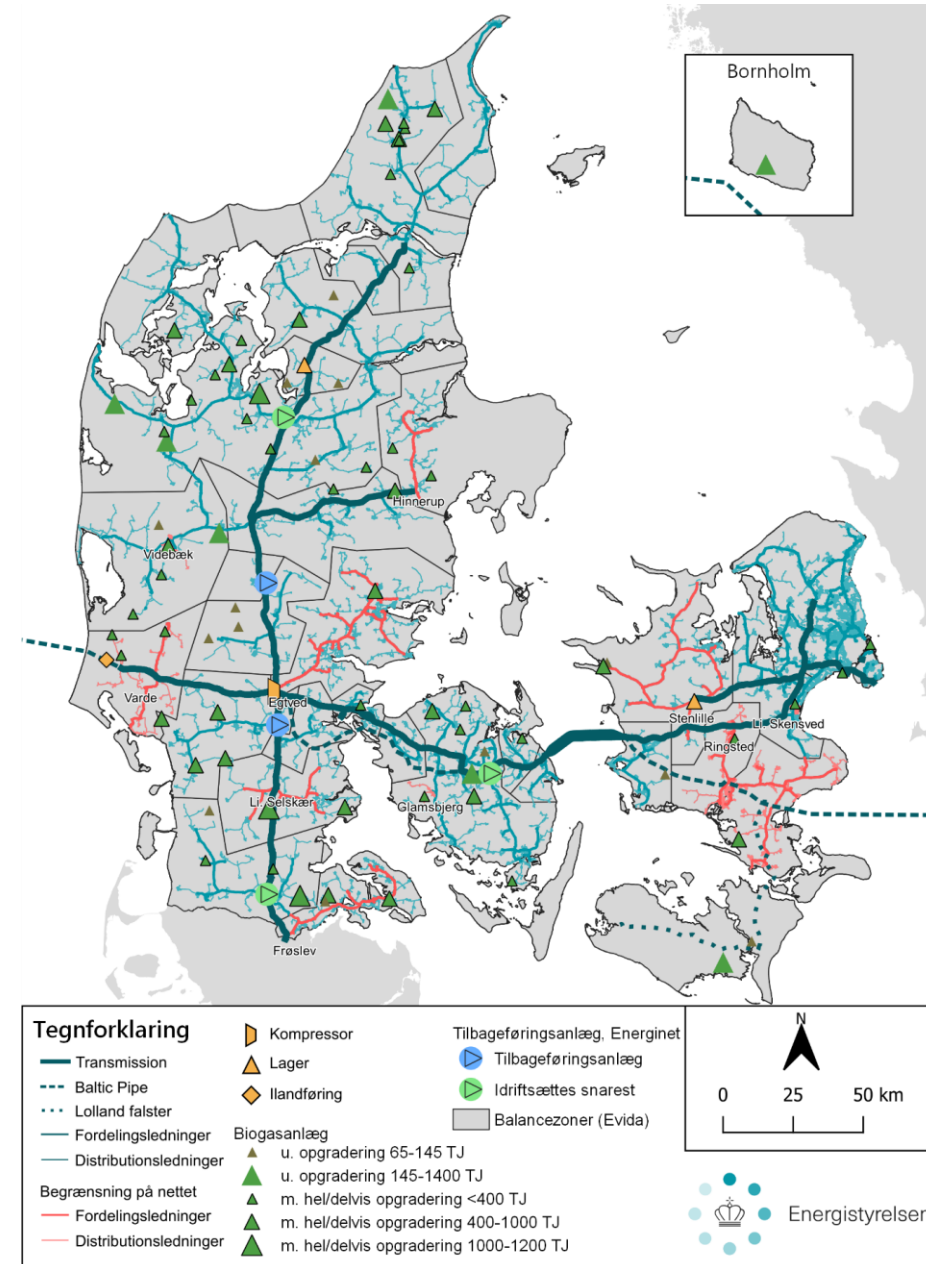


# Challenges and options

## Adapting the gas-grid(s)



- Adjusting the system to new flow patterns with decentral RE-gas production
- Modify the system for new gasses to transport
- Modify the support scheme
- Blending obligations
- Tax exemptions
- Sustainability



# Future considerations

- RED III
- Exchange bio with heat pumps
- Energy parks and green fuels
- Biorefinery and prioritising biomass use





Questions?

Thank you

